

BIOS vs UEFI Comparison

Category	BIOS	UEFI
Definition	BIOS (Basic Input/Output System) is older firmware that initializes hardware and starts the operating system.	UEFI (Unified Extensible Firmware Interface) is a modern firmware interface designed to replace traditional BIOS.
Boot Process	Uses a traditional boot process where BIOS initializes hardware and loads the boot code from the disk's MBR.	Initializes hardware and loads a boot manager from the EFI System Partition (ESP).
Maximum Disk Support	Traditionally limited by the MBR partitioning scheme to about 2.2 TB for a boot disk.	Works with GPT, allowing support for disks much larger than 2 TB. GPT's theoretical limit is far beyond today's typical disk sizes.
Partition Style	Commonly uses MBR (Master Boot Record) .	Commonly uses GPT (GUID Partition Table) . A UEFI Windows boot disk must use GPT.
Security Features	Has limited built-in boot security and does not provide UEFI's Secure Boot mechanism.	Supports Secure Boot , which verifies trusted boot software before allowing the system to start.
Speed	Generally has a slower and more limited boot process.	Generally provides faster boot and resume times.
Advantages	Simple, widely compatible with older hardware and operating systems, and useful for legacy systems.	Faster startup, supports large disks and more partitions, provides modern security features, and offers greater flexibility.
Disadvantages	Limited by older technology, MBR restrictions, and fewer modern security features.	Can be more complex to configure and may have compatibility issues with very old operating systems or hardware.

Category	BIOS	UEFI
Modern Usage	Mostly used for legacy computers and compatibility with older systems.	Used on most modern computers and is the preferred firmware standard for current systems.

Why UEFI Has Largely Replaced BIOS

BIOS and UEFI are both firmware interfaces that help a computer start and load an operating system. However, BIOS is an older technology designed for the early generation of personal computers. As computer hardware developed, the limitations of BIOS became more noticeable. UEFI was developed as a modern replacement for the traditional BIOS interface. Microsoft describes UEFI as a replacement for the older BIOS firmware interface.

One major advantage of UEFI is its support for **larger storage devices**. Traditional BIOS commonly works with the MBR partitioning system, which has a practical limit of approximately 2.2 TB for addressing a boot disk. UEFI works with GPT, which supports much larger storage capacities and allows more partitions. GPT can support up to 128 partitions in a standard Windows UEFI configuration.

UEFI also provides important **security improvements**. One of its most important features is Secure Boot. Secure Boot checks the digital signatures of boot software before allowing it to run. This helps prevent unauthorized or malicious software, such as certain bootkits and rootkits, from loading during the startup process.

Another benefit is **faster startup and resume times**. UEFI provides a more modern pre-boot environment and is designed to work efficiently with current hardware. Microsoft specifically identifies faster boot and resume times as benefits of UEFI.

UEFI is also more flexible than traditional BIOS. It can provide a standardized environment for firmware drivers and boot applications, making it better suited to modern operating systems and hardware. Modern Windows systems are designed around UEFI, GPT, Secure Boot, and related security technologies.

In conclusion, **UEFI has largely replaced BIOS because it removes important limitations of the older BIOS design while providing better storage support, faster booting, improved security, and greater compatibility with modern computer hardware and operating systems**. BIOS is still useful for older computers and legacy compatibility, but UEFI is the preferred firmware technology for modern computers.

References

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